1

Lesson: Introduction to C

CSE 4107 : Structured Programming I

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2

Origins of C

- A byproduct of UNIX operating systems
- Developed at Bell Laboratories by Ken Thompson, Dennis Ritchie and others
- UNIX was written in Assembly language
- Debugging complications inspired Thompson to write a new language - B
- B was based on BCPL (a systems programming language)

3

Origins of C

- By 1971, Ritchie began to develop an extended version of B, which he named **NB** (“New B”)
- As the language diverged from B, he changed its name to C
- UNIX was rewritten in C in 1973

7

C based languages

- ***C++*** includes all the features of C, but adds feature support for object oriented programming
- ***Java*** based on C++ and therefore inherits many features
- ***C#*** is more recent language derived from Java and C++
- ***Perl*** another scripting language, adopted many features of C
- ***JavaScript*** uses syntax influenced by C

8

Properties of C

- Low level language
  - Provides access to machine level concepts
- Comparatively small
  - C keeps itself limited by relying heavily on a library of standard functions
- Permissive
  - Provides more freedom compared to other higher level languages (which is also more risky)